

AI Recommendation Logic

Project Documentation

Name	Shruti
Internship	Decode Lab internship
Project	AI Recommendation Logic
Technology	Python

1. Introduction

This project is based on the DecodeLabs Industrial Training Kit, Artificial Intelligence Project 3. The project focuses on AI Recommendation Logic and introduces recommendation systems through preference matching and similarity-based logic. The supplied brief emphasizes pattern alignment rather than random suggestions.

2. Project Goal

The goal is to create a simple recommendation system based on user preferences. The system accepts user interests, compares them with item attributes, and displays the most relevant recommendations.

3. Key Requirements

- Take user input in the form of choices or interests.
- Match user preferences using logic or similarity.
- Display recommended items.

4. Approach

A simple content-based approach is used. Each item is represented by a set of attributes or tags. The user's selected interests are compared with those tags. A similarity score is calculated as the number of matching interests divided by the number of user interests. Items with higher scores are shown first.

5. Algorithm

- Store a small catalog of items with descriptive tags.
- Ask the user to enter interests separated by commas.
- Normalize the entered interests.
- Calculate the overlap between user interests and each item's tags.
- Rank items by their similarity score.
- Display the recommendations and their scores.

6. Example

If a user selects interests such as Python and AI, items containing both tags receive a higher score than items containing only one of them. This demonstrates the pattern-matching principle described in the project brief.

7. Skills Demonstrated

- Logic building
- Pattern matching
- Basic recommendation concepts
- Python programming
- User input handling

8. Conclusion

The AI Recommendation Logic project demonstrates how raw user preferences can be converted into relevant recommendations through straightforward algorithmic matching. It provides a foundation for understanding recommendation systems before moving toward more complex machine-learning approaches.

9. Reference

DecodeLabs Industrial Training Kit, Batch 2026, Artificial Intelligence — Project 3: AI Recommendation Logic.